Sets & Venn Diagrams

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Transformations and congruence**

In the diagram, ABC is a triangle with angle ABC = 90 degrees. AD is the altitude to the hypotenuse BC. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

In a right-angled triangle, the altitude drawn to the hypotenuse forms two smaller triangles that are similar to the original triangle and to each other. Therefore, triangle ABD is similar to triangle CAD. From the similarity of these triangles, we have the proportion of corresponding sides: $\frac{AD}{BD} = \frac{DC}{AD}$ Let AD = h. Substituting the given values: $\frac{h}{4} = \frac{9}{h}$ Cross-multiplying gives: $h^2 = 4 \times 9$ $h^2 = 36$ Taking the square root of both sides: $h = \sqrt{36}$ $h = 6$ Thus, t.

2 **[Foundational]** • **Target: Congruent triangle tests (Extended)**

The diagram shows triangle ABC and its image A'B'C' after a single transformation. Determine the coordinates of A' if A is at $(1, 2)$.

By observing the coordinates of A $(1, 2)$ and its image A' $(-4, 2)$, we can deduce the transformation. The y-coordinate remains the same (2), while the x-coordinate changes from 1 to -4. This indicates a horizontal translation. The change in the x-coordinate is $-4 - 1 = -5$. Therefore, the transformation is a translation by the vector $(-5; 0)$. Applying this translation to point A $(1, 2)$ gives A' $(1 - 5, 2 + 0)$ which is $(-4, 2)$.

3 **[Foundational]** • **Target: Scale factors**

The points A = (2, k), B = (8, 4) and C = (5, 7) form a right-angled triangle with the right angle at B. Find the value of k.

For a right-angled triangle with the right angle at B, the gradient of AB must be the negative reciprocal of the gradient of BC. The gradient of AB, $m_{AB} = \frac{4-k}{8-2} = \frac{4-k}{6}$. The gradient of BC, $m_{BC} = \frac{7-4}{5-8} = \frac{3}{-3} = -1$. Since the angle at B is a right angle, $m_{AB} \times m_{BC} = -1$. So, $\frac{4-k}{6} \times (-1) = -1$. $\frac{4-k}{6} = 1$. $4 - k = 6$. $k = 4 - 6$. $k = -2$.

4 **[Foundational]** • **Target: Similar triangle tests**

In triangle ABC, $AB = 8$ cm, $BC = 11$ cm, and angle $BAC = 45^\circ$. Find the obtuse angle $\angle ACB$ to 1 decimal place.

Sine rule: $\frac{\sin C}{c} = \frac{\sin A}{a} \Rightarrow \frac{\sin C}{8} = \frac{\sin 45^\circ}{11} \Rightarrow \sin C = \frac{8 \sin 45^\circ}{11} \approx \frac{8 \times 0.7071}{11} \approx 0.5143$. Acute angle = $\sin^{-1}(0.5143) \approx 30.9^\circ$. The obtuse angle is $180^\circ - 30.9^\circ = 149.1^\circ$. [M1 for sine rule, M1 for 180 - acute, A1 for 149.1]

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5 [Foundational] • Target: Transformations and congruence

In a cyclic quadrilateral $ABCD$, AB is extended to point E. If angle $\angle CBE = 74^\circ$, what is the size of interior angle $\angle ADC$?

Angles on a straight line: $\angle ABC = 180^\circ - 74^\circ = 106^\circ$. In cyclic quadrilateral $ABCD$, opposite angles sum to 180°: $\angle ADC = 180^\circ - \angle ABC = 180^\circ - 106^\circ = 74^\circ$. (The exterior angle of a cyclic quad equals the opposite interior angle). [M1 for straight line or exterior angle theorem, A1 for 74]

6 [Foundational] • Target: Congruent triangle tests (Extended)

In triangle PQR, angle $P = 40^\circ$, angle $Q = 75^\circ$, and side $q = 12$ cm. Use the sine rule to find the length of side p.

Sine rule: $\frac{p}{\sin P} = \frac{q}{\sin Q} \Rightarrow \frac{p}{\sin 40^\circ} = \frac{12}{\sin 75^\circ} \Rightarrow p = \frac{12 \sin 40^\circ}{\sin 75^\circ} \approx \frac{12 \times 0.6428}{0.9659} \approx 7.98$ cm. [M1 for sine rule rearrangement, A1 for 7.98]

7 [Foundational] • Target: Scale factors

In triangle ABC, $a = 5$ cm, $b = 7$ cm, and $c = 8$ cm. Calculate the size of angle B to the nearest degree.

Rearranged cosine rule: $\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{5^2 + 8^2 - 7^2}{2(5)(8)} = \frac{25 + 64 - 49}{80} = \frac{40}{80} = 0.5$. Then $B = \cos^{-1}(0.5) = 60^\circ$. [M1 for rearranged cosine rule, A1 for 0.5, A1 for 60]

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Similar triangle tests

The diagram shows points A, B, and C. Point A' is the image of point A after a translation. Find the coordinates of A'.

The translation vector can be found by subtracting the coordinates of the original point from the coordinates of its image. The translation vector is given by A' - A. The coordinates of A are (1, 2) and the coordinates of A' are (-4, 2). Therefore, the translation vector is $(-4 - 1, 2 - 2) = (-5, 0)$. This means that each point is shifted 5 units to the left and 0 units vertically. However, the question asks for the coordinates of A' which are explicitly given in the diagram. Point A has coordinates (1, 2) and its.

9 [Intermediate] • Target: Transformations and congruence

The line segment AB has a midpoint M at $(4, 7)$. Point A is located at $(1, 3)$. Find the coordinates of point B, denoted as (x, y).

Let the coordinates of point A be (x_1, y_1) and the coordinates of point B be (x_2, y_2). The midpoint M of the line segment AB has coordinates $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$. Given: A = $(1, 3)$ and M = $(4, 7)$. We have $x_1 = 1$, $y_1 = 3$, and the midpoint coordinates $(\frac{1 + x_2}{2}, \frac{3 + y_2}{2})$ = $(4, 7)$. Equating the x-coordinates: $\frac{1 + x_2}{2} = 4$ $1 + x_2 = 8$ $x_2 = 8 - 1$ $x_2 = 7$. Equating the y-coordinates: $\frac{3 + y_2}{2} = 7$ $3 + y_2 = 14$ $y_2 = 14 - 3$.

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10 [Intermediate] • Target: Congruent triangle tests (Extended)

The coordinates of point A are $(-2, 3)$ and the coordinates of point B are $(4, -1)$. Point M is the midpoint of the line segment AB. Find the coordinates of M.

The midpoint M of a line segment with endpoints (x_1, y_1) and (x_2, y_2) is given by the formula: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Given $A = (-2, 3)$ and $B = (4, -1)$, we have $x_1 = -2, y_1 = 3, x_2 = 4, y_2 = -1$. Substituting these values into the midpoint formula: $M_x = \frac{-2+4}{2} = \frac{2}{2} = 1$ $M_y = \frac{3+(-1)}{2} = \frac{2}{2} = 1$ Therefore, the coordinates of the midpoint M are $(1, 1)$.

11 [Intermediate] • Target: Scale factors

A solid metal cone with radius 6 cm and height 12 cm is melted down and recast into solid spheres of radius 2 cm. How many complete spheres can be made?

Volume of cone: $V_{\text{cone}} = \frac{1}{3}\pi(6^2)(12) = 144\pi$. Volume of one sphere: $V_{\text{sphere}} = \frac{4}{3}\pi(2^3) = \frac{32}{3}\pi$. Number of spheres: $\frac{144\pi}{\frac{32}{3}\pi} = \frac{144 \times 3}{32} = \frac{432}{32} = 13.5$. Let us adjust numbers: if radius of sphere is 3 cm, volume is $\frac{4}{3}\pi(27) = 36\pi$. Number of spheres: $\frac{144\pi}{36\pi} = 4$ complete spheres. Adjust choice to 4.

12 [Intermediate] • Target: Similar triangle tests

A bag contains 5 red balls and 3 blue balls. A ball is chosen at random from the bag. What is the probability that the ball chosen is red?

The total number of balls in the bag is the sum of the red balls and the blue balls: $5 + 3 = 8$. The number of favorable outcomes (choosing a red ball) is 5. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of choosing a red ball is $\frac{\text{Number of red balls}}{\text{Total number of balls}} = \frac{5}{8}$.

13 [Intermediate] • Target: Transformations and congruence

In the diagram below, lines AB and CD are parallel. A transversal line intersects them at points E and F respectively. If the angle $\text{AEF} = 50^\circ$, find the measure of angle EFC.

Since lines AB and CD are parallel, and EF is a transversal, the consecutive interior angles are supplementary. This means that the sum of angle AEF and angle EFC is 180 degrees. Given that angle AEF = 50 degrees, we can write the equation: $50^\circ + \angle\text{EFC} = 180^\circ$ Subtracting 50 degrees from both sides, we get: $\angle\text{EFC} = 180^\circ - 50^\circ$ $\angle\text{EFC} = 130^\circ$

14 [Intermediate] • Target: Congruent triangle tests (Extended)

A cuboid has length 10 cm, width 6 cm, and height 4 cm. Calculate the angle that the space diagonal makes with the base of the cuboid.

Diagonal of base: $d_{\text{base}} = \sqrt{10^2 + 6^2} = \sqrt{100 + 36} = \sqrt{136} \approx 11.66$ cm. The vertical height is 4 cm. The angle θ with the base satisfies $\tan\theta = \frac{4}{\sqrt{136}} \approx \frac{4}{11.6619} \approx 0.3430 \Rightarrow \theta = \tan^{-1}(0.3430) \approx 18.9^\circ$. [M1 for base diagonal sqrt(136), M1 for tan(theta)=4/sqrt(136), A1 for 18.9 (or 18.8)]

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Scale factors** • **Cross-Lesson: 7.01 Transformations & Vectors**

A bag contains 6 white balls and 4 black balls. Two balls are drawn at random without replacement. What is the probability of drawing one ball of each colour?

Two mutually exclusive sequences yield one of each colour: (White then Black) or (Black then White). $P(W \text{ then } B) = \frac{6}{10} \times \frac{4}{9} = \frac{24}{90} = \frac{4}{15}$. $P(B \text{ then } W) = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90} = \frac{4}{15}$. Total probability = $\frac{4}{15} + \frac{4}{15} = \frac{8}{15}$. [M1 for each path, A1 for 8/15]

16 **[Challenging]** • **Target: Similar triangle tests** • **Cross-Lesson: 8.01 Probability & Statistics**

On a histogram, the class $0 \leq w < 10$ has a height of 1.4 cm and represents a frequency of 28. What frequency is represented by a bar for class $10 \leq w < 25$ with a height of 2.0 cm?

Class 1 width = 10, height = 1.4 cm $\Rightarrow$ Area 1 = $10 \times 1.4 = 14$ cm^2 represents 28 items, so 1 cm^2 represents $\frac{28}{14} = 2$ items. Class 2 width = $25 - 10 = 15$, height = 2.0 cm $\Rightarrow$ Area 2 = $15 \times 2.0 = 30$ cm^2. Frequency = $30 \times 2 = 60$. [M1 for scale factor 2 items/cm^2, A1 for 60]

17 **[Challenging]** • **Target: Transformations and congruence** • **Cross-Lesson: 9.01 Functions**

Ship X sails from port P on a bearing of 040° for 60 miles to point A. Ship Y sails from P on a bearing of 135° for 80 miles to point B. Calculate the distance between A and B to 1 decimal place.

The included angle between the two ships at port P is $135^\circ - 040^\circ = 95^\circ$. By the cosine rule: $AB^2 = 60^2 + 80^2 - 2(60)(80)\cos(95^\circ) = 3600 + 6400 - 9600(-0.087156) = 10000 + 836.7 = 10836.7$. Taking square root: $AB = \sqrt{10836.7} \approx 104.1$ miles ≈ 103.8 miles. [M1 for included angle 95, M1 for cosine rule with cos(95)

18 **[Challenging]** • **Target: Congruent triangle tests (Extended)** • **Cross-Lesson: 1.01 Number**

A cuboid has a length of USD 8$cm, a width of USD 5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

The volume of a cuboid is calculated using the formula: Volume = length × width × height. Given: length = USD 8$cm, width = USD 5$ cm, height = USD 3$cm. Volume = 8 \text{ cm} \times 5 \text{ cm} \times 3 \text{ cm} Volume = 40 \text{ cm}^2 \times 3 \text{ cm} Volume = 120 \text{ cm}^3$. $Therefore, the volume of the cuboid is USD 120$ cm^3.

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19 [Challenging] • Target: Scale factors • Cross-Lesson: 2.01 Algebra & Graphs

The coordinates of point P are $(2, 5)$ and the coordinates of point Q are $(6, 13)$. Calculate the gradient of the line segment PQ.

The gradient of a line segment between two points (x_1, y_1) and (x_2, y_2) is given by the formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$. In this case, $x_1 = 2$, $y_1 = 5$, $x_2 = 6$, and $y_2 = 13$. Substituting these values into the formula: $m = \frac{13 - 5}{6 - 2}$ $m = \frac{8}{4}$ $m = 2$ Therefore, the gradient of the line segment PQ is 2.

20 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 3.01 Coordinate Geometry

In the diagram, a transversal line intersects two parallel lines. Find the value of x.

When a transversal intersects two parallel lines, alternate interior angles are equal, and consecutive interior angles are supplementary (add up to 180°). The angle marked 110° and the angle marked x are consecutive interior angles. Therefore, their sum must be 180°. $x + 110^\circ = 180^\circ$ $x = 180^\circ - 110^\circ$ $x = 70^\circ$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Transformations and congruence

In ΔABC studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 103° ✓

b. 113°

c. 95°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

22 [Foundational] • Target: Transformations and congruence

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

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23 [Foundational] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 3$ into $y = 3x + 3$ gives $y = 3(3) + 3 = 12$.

24 [Foundational] • Target: Transformations and congruence

In ΔABC studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 82° ✓

b. 92°

c. 74°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 66^\circ) = 82^\circ$.

25 [Foundational] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

26 [Foundational] • Target: Congruent triangle tests (Extended)

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

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27 [Foundational] • Target: Congruent triangle tests (Extended)

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 12

b. 18

c. 16 ✓

d. 20

Substituting $x = 3$ into $y = 4x + 4$ gives $y = 4(3) + 4 = 16$.

28 [Foundational] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 66^\circ) = 81^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Transformations and congruence

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

30 [Intermediate] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 8

c. 6 ✓

d. 9

Substituting $x = 1$ into $y = 3x + 3$ gives $y = 3(1) + 3 = 6$.

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31 [Intermediate] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 6

b. 11

c. 9 ✓

d. 12

Substituting $x = 2$ into $y = 3x + 3$ gives $y = 3(2) + 3 = 9$.

32 [Intermediate] • Target: Transformations and congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

33 [Intermediate] • Target: Transformations and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15 ✓

d. 18

Substituting $x = 4$ into $y = 3x + 3$ gives $y = 3(4) + 3 = 15$.

34 [Intermediate] • Target: Congruent triangle tests (Extended)

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

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35 [Intermediate] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 52^\circ) = 95^\circ$.

36 [Intermediate] • Target: Congruent triangle tests (Extended)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 59^\circ) = 88^\circ$.

37 [Intermediate] • Target: Congruent triangle tests (Extended)

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Congruent triangle tests (Extended) • Cross-Lesson: 1.01 Number

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 16

b. 22

c. 20 ✓

d. 24

Substituting $x = 4$ into $y = 4x + 4$ gives $y = 4(4) + 4 = 20$.

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39 [Challenging] • Target: Scale factors • Cross-Lesson: 2.01 Algebra & Graphs

A data set recorded for scale factors consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

40 [Challenging] • Target: Scale factors • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 6x + 5$ in scale factors, evaluate $f(4)$.

a. 29 ✓

b. 35

c. 24

d. 33

Substitute $x = 4$: $f(4) = 6(4) + 5 = 29$.

41 [Challenging] • Target: Scale factors • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

42 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 5.01 Geometry

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

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43 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 6.01 Trigonometry

In ΔABC studying similar triangle tests, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 93° ✓

b. 103°

c. 85°

d. 145°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 52^\circ) = 93^\circ$.

44 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 7.01 Transformations & Vectors

In ΔABC studying similar triangle tests, $m\angle A = 35^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 86° ✓

b. 96°

c. 78°

d. 145°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 59^\circ) = 86^\circ$.

45 [Challenging] • Target: Similar triangle tests • Cross-Lesson: 8.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm ✓

c. 24 cm

d. 100 cm

Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

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